

RepoScope

See your repo. Understand it. Track it.

v3.0 - from understanding to building

Built with IBM Bob + Z Code · lablab.ai · Devpost

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SESSION LOG

S1

Backend core

FastAPI server · graph_builder · GitHub client · SSE streaming

S2

Frontend workspace

React + React Flow graph UI · chat panel · code tours

S3

v3.0 features

PR bot · student mode · voice tours · security scan · scaffold

● IBM Bob

● Z Code

pair-programmed end to end

```
● ● ●  ibm bob · session 3

> bob: add a security scan endpoint
created routes/security.py
wrote 3 regex detectors
tests green ✓
session exported → docs/bob-sessions/s3.md
```

github.com/<your-handle>/reposcope/tree/main/docs/bob-sessions

Demo repos.

two real repos — one knowledge graph each

Verdict

Python · FastAPI

```
graph TD; github[github] -- green arrow --> api[api]; ui[ui] -- black arrow --> api; api -- black arrow --> db[db]; api -- black arrow --> llm[llm]; llm -- black arrow --> sse[sse]; api -- black arrow --> auth[auth]; auth -- black arrow --> sse
```

The diagram shows the architecture of the Verdict project. It features a central 'api' node. 'github' and 'ui' both point to 'api' via black arrows, with a green arrow specifically highlighting the connection from 'github'. 'api' then branches out to 'db', 'llm', and 'auth'. 'llm' points to 'sse', and 'auth' also points to 'sse'.

- Architecture graph + impact analysis demo
- Scaffold example — backend/main.py starter kit

StudyRot

student study app

```
graph TD; ui[ui] -- black arrow --> db[db]; state[state] -- black arrow --> db; state -- green arrow --> tour[tour]; db -- black arrow --> quiz[quiz]; tour -- black arrow --> quiz; tour -- black arrow --> tts[tts]
```

The diagram shows the architecture of the StudyRot project. It features a central 'db' node. 'ui' and 'state' both point to 'db' via black arrows. 'state' also points to 'tour' via a green arrow. 'db' points to 'quiz', and 'tour' also points to 'quiz'. Finally, 'tour' points to 'tts'.

- Code tours + voice narration demo
- Student Onboarding Mode — Row 0–5 path

both repos scanned live in the demo video

Live right now.

● ALL SYSTEMS LIVE



Live app

repo-scope-nu.vercel.app

● deployed · Vercel



API health

reposope.onrender.com/api/health

● deployed · Render · 200 OK



Demo video

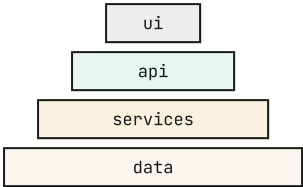
youtu.be/W82roIETJJ8

● 3-minute walkthrough

Paste any public GitHub URL — the graph renders in seconds. Try it during Q&A.

Core features.

01



Architecture Graph
Layered map of the whole repo — client to compute.

02



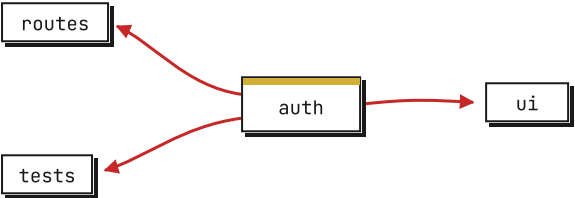
Q&A with citations
Answers cite files, functions, and commits.

03



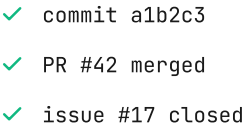
Code Tours
Guided walkthroughs of any module.

04



Impact Analysis
Change a function — see the blast radius.

05



Tracking
Commits, PRs, issues, releases — on demand.

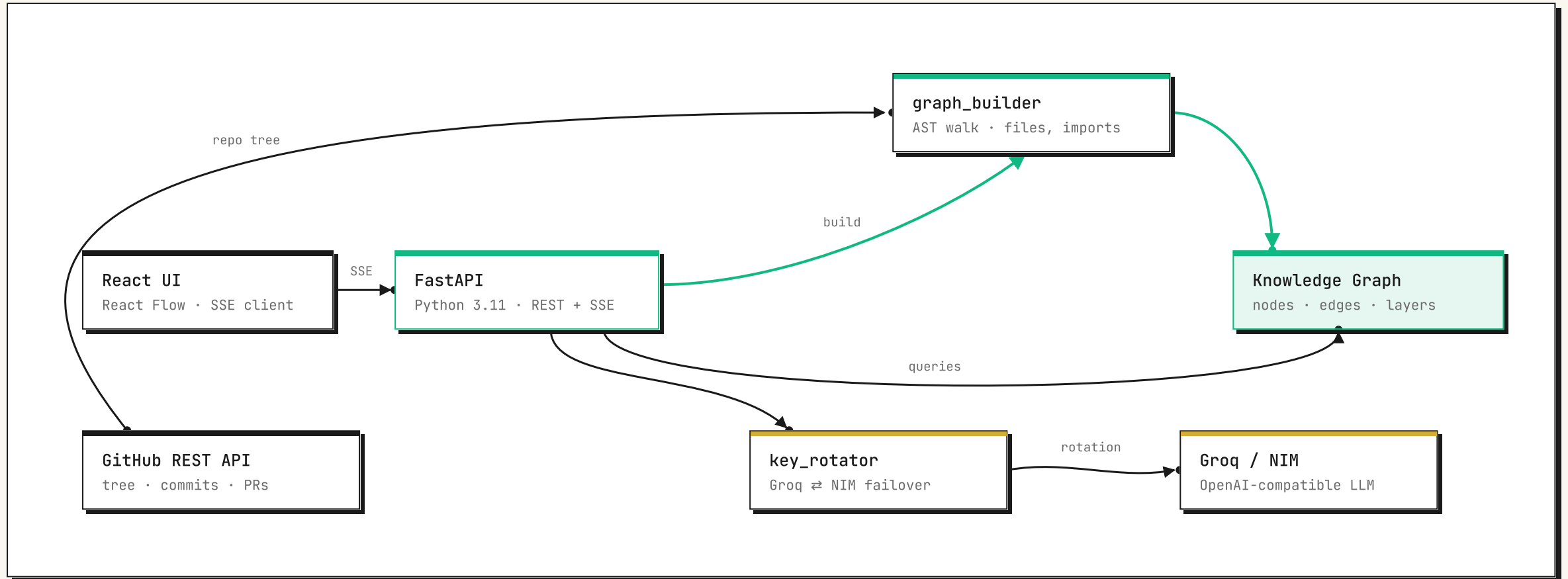
06



Reverse Prompt
Any repo → an agent-ready rebuild prompt.

Architecture.

how a GitHub URL becomes a living knowledge graph



■ graph_builder – one pass per repo, AST to edges

■ key_rotator – Groq primary, NIM fallback

■ every feature is a query on the same graph

NEW IN 3.0

5 features · 1 graph

From Understanding to Building.

v2 read your repo. v3.0 helps you ship to it.

01



PR Bot

impact review on every PR

02



Student Onboarding

guided path to a first PR

03



Voice Tours

narrated graph walkthroughs

04



Security Scan

secrets · CVEs · breaking bumps

05

HERO



Scaffold

repo → starter kit in 5 sec

each feature → one deep-dive slide next

PR Bot.

impact review lands on every pull request

RS

RepoScope-Bot

BOT

commented • now

Impact analysis for this PR

RISK SCORE

72

/ 100

MODERATE

IMPACTED FILES

services/auth.py	+12 -4
routes/api.py	+8 -2
tests/test_auth.py	+21 -0

impact.png

blast radius

```
graph LR; A[services/auth.py changed] --> B[routes/api.py impacted]; A --> C[tests/test_auth.py impacted]; A --> D[db/session.py impacted];
```

- **POST /api/impact/pr**
one endpoint — fired automatically on every opened PR
 - **Reuses impact_analyzer**
the exact engine behind the in-app impact panel — zero duplicated logic
 - **.github/workflows/pr-impact.yml**
a GitHub Action workflow — install, comment, done
 - **Impact panel preview button**
one click from the comment jumps into the full interactive panel
- IMPACT

Prevents breaking changes in CI.

Student Onboarding Mode.

from lost → first pull request

StudyRot • workspace

Student Mode ☒

GRAPH RANKING

rank = incoming × 2 + outgoing

Row 0

Row 1

Row 2

Row 3

Row 4

Row 5

← Good First Issues

LEVEL PATH

Level 1 — Repo basics

■■■■ quiz 3/3 ✓

Level 2 — Core modules

■■■□ quiz 2/3

Level 3 — First contribution

Locked

- **Graph ranking — incoming × 2 + outgoing**
every file lands on a difficulty row from 0 to 5
- **Row 4–5 = Good First Issues**
gentle, well-scoped entry points surfaced automatically
- **3 quizzes per level**
understanding is verified before the path unlocks

IMPACT

Onboarding: days → 20 minutes.

● ML Empowerment • education track

▶ Play Tour

narration: browser TTS

NOW PLAYING

Stop 2 / 6 — services/auth.py

main.py

sse.py

routes/api.py

services/auth.py
stop 2

llm/groq.py

db/models.py

```
graph TD; main.py --> auth.py; sse.py --> auth.py; routes/api.py --> auth.py; auth.py --> llm/groq.py; auth.py --> db/models.py;
```

- **Web Speech API · `speechSynthesis`**
built into every modern browser — zero cost, zero cloud calls
- **Auto-pan sync**
the graph camera follows the narration stop by stop
- **Accessibility first**
blind and low-vision students can explore a repo by ear
- **Clones the Aster winner**
the voice-tour experience judges already awarded — now on any repo

A repo tour you can take with your eyes closed.

Security Scan.

three flags. zero surprises in production

Graph	Chat	Tours	Security	Settings
Run scan				3 findings
BREAKING pydantic 1.10 → 2.0 • major bump				pin or migrate
SECRET API key regex • services/llm.py:42				rotate key
CVE requests 2.28.0 • known advisory				upgrade

- POST /api/security/scan
one endpoint returns the full flag list
- API-key regex detection
sk-, ghp_, AKIA... patterns flagged on sight
- .env leak detection
secrets committed to git history are caught
- Major bump = BREAKING
dependency major-version jumps flagged red before CI does

Nord Security track

IMPACT

Catch it in the tab — not in the incident channel.

Scaffold.

From Repo to Starter Kit **in 5 Sec** CodeCrafters-style starter kits

BEFORE — REVERSE PROMPT

```
# Reverse Build Prompt — verdict
Rebuild this repo: FastAPI backend,
3 services, Postgres store, SSE streaming.
You are an agent — rebuild it.
```

text a human still has to interpret

5 sec

AFTER — SCAFFOLD MODAL

Scaffold — verdict

backend/

main.py

requirements.txt

README.md

```
1 from fastapi import FastAPI
2 app = FastAPI()
3 @app.post("/predict")
4 def predict():
5     # TODO: implement predict
```

Download ZIP

Open in StackBlitz

verdict — starter kit

Tested. All green.

every command below ran against the deployed build

```
○ ○ ○ reposcope — zsh

$ pytest
✓ all tests passed

$ npm run build
✓ build complete — 0 errors

$ curl reposcope.onrender.com/api/health
{ "status": "ok" }

$ curl -I repo-scope-nu.vercel.app
HTTP/2 200

$ █
```

✓	pytest	green
✓	curl	green
✓	npm run build	green
✓	/api/health	green

UI CLICKS

All 5 features tested via UI clicks

- ✓ PR Bot
- ✓ Student Mode
- ✓ Voice Tours
- ✓ Security Scan
- ✓ Scaffold

Built for the judges.

two tracks, one live demo

● ML EMPOWERMENT · education track

■ Student Onboarding Mode

onboarding drops from days to 20 minutes

■ Voice Code Tours

accessible by design — blind students explore by ear

■ Quizzes + levels

a real learning loop, not a dashboard

● LOVHACK · automation track

■ PR Bot

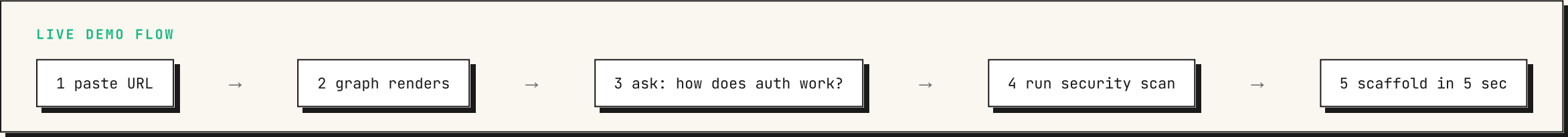
breaking changes blocked before CI

■ Scaffold

any repo → starter kit in 5 seconds

■ Devtool wow

one graph under eleven features — the demo writes itself



Tech stack.



Open source. MIT.

fork it, ship it, extend it

 LICENSE MIT

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to any person obtaining a copy of this software...

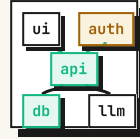
...use, copy, modify, merge, publish, distribute,
sublicense, and/or sell copies...

- free to fork

- free to ship

- free to extend

github.com/<your-handle>/reposcope/blob/main/LICENSE



Try RepoScope

See your repo. Understand it. Track it.



GitHub

github.com/adityapatel5912/RepoScope



Live app

repo-scope-nu.vercel.app



Video

youtu.be/W82roIETJJ8

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